

IT Security 2

Unit 5 – Security Zones

Instructor: Brian Krause

Announcements



YOUR VOICE MATTERS!

**New Mid-State Students:
Check Your Email!**

Participate in the **SENSE** Survey

What: Survey of Entering Student Engagement (SENSE) 
When: Sept. 23 - Nov7, 2025 
Who: New Fall 2025 Students (18+) 

Your feedback helps us make Mid-State even better!

Questions? Email: Sheila.cover@mstc.edu

Sign up for the Etiquette Luncheon!

Essential Details LLC will be on campus next month to host this fantastic, **FREE** event.



During the Perfectly Polished Professional session on etiquette, attendees will skills for

- professional meetings
- best practices during mealtimes
- networking tips

Spots are limited!

Click [HERE](#) to register



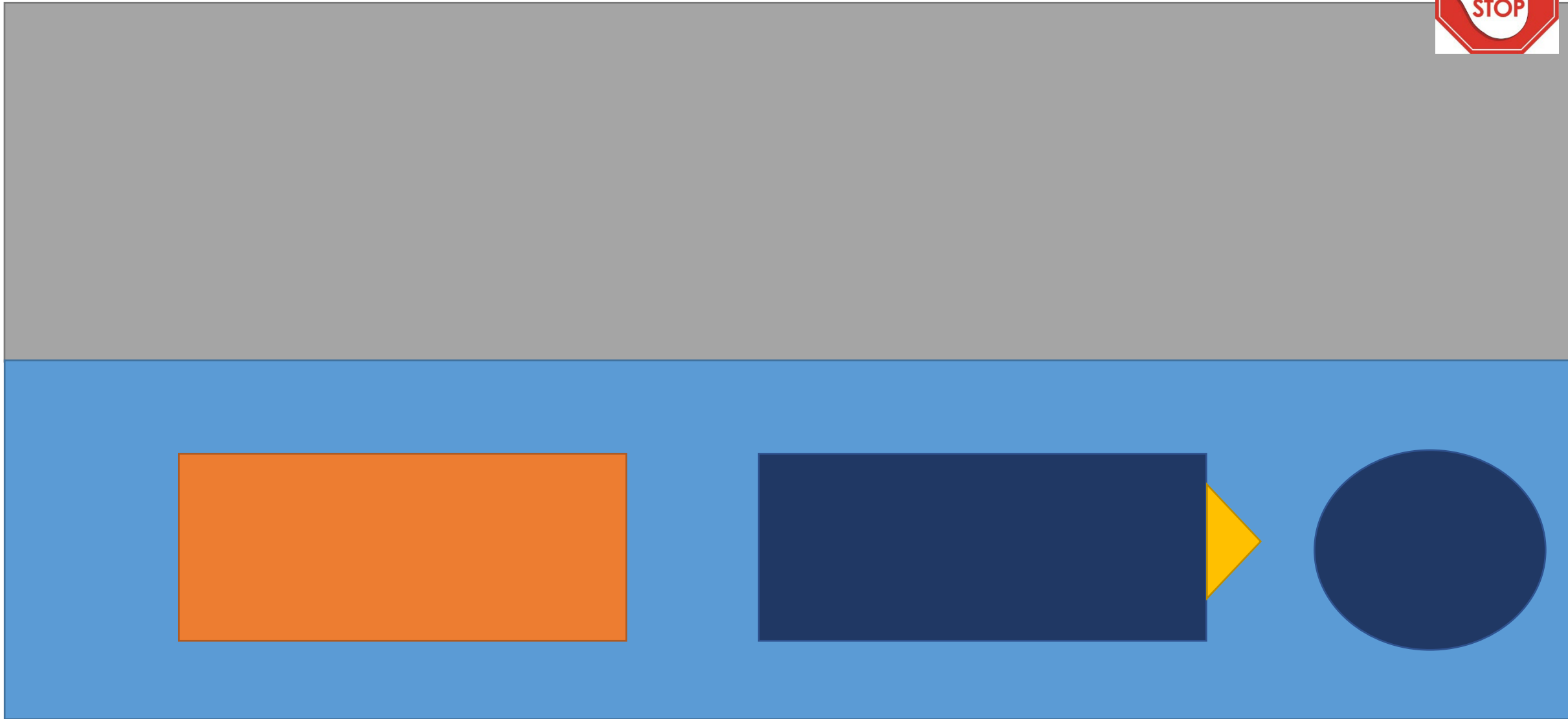
Scan to register

 Thursday, October 9
11:00 am. - 1:00 p.m.
Stevens Point Downtown Campus
LEAD Center

Security Zones

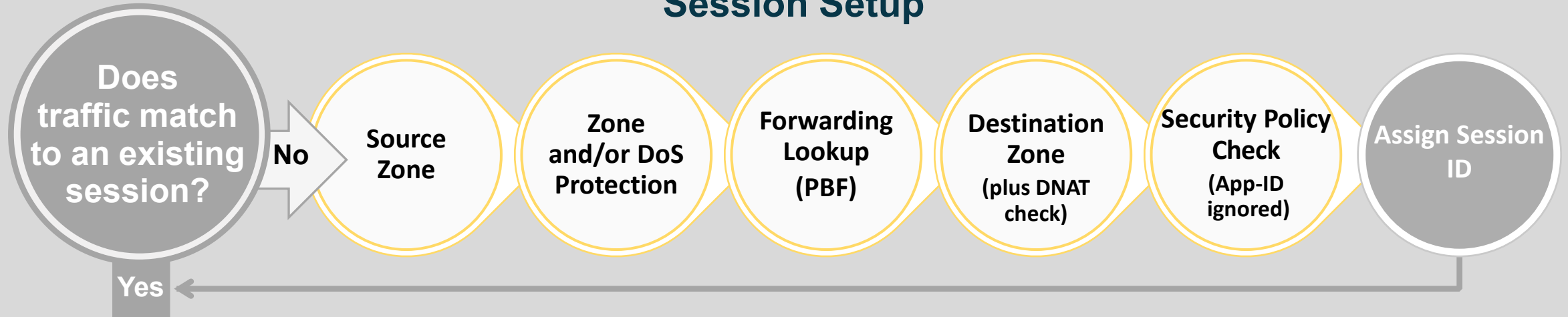
Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

Flow Logic of the Next-Generation Firewall

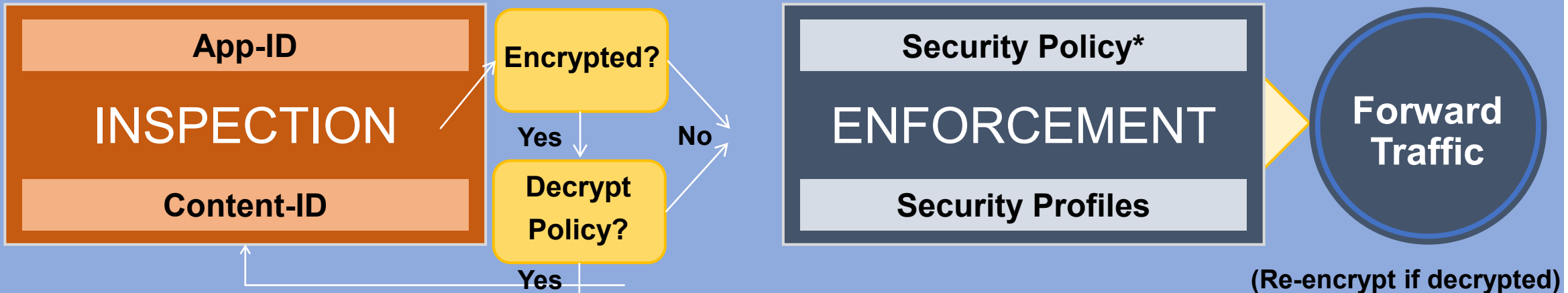


Flow Logic of the Next-Generation Firewall

Session Setup

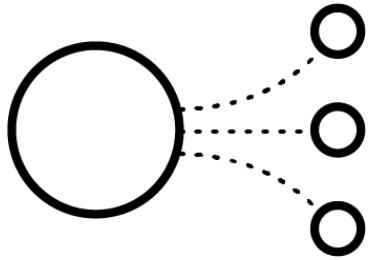


Inspection and Enforcement



* Policy check relies on pre-NAT IP addresses

CONNECTING THE FIREWALL TO PRODUCTION NETWORKS WITH SECURITY ZONES



NETWORK CONNECTIVITY IS NOT A LUXURY, IT IS A NECESSITY

- Security Zones Overview
- Network interfaces and security zones
- Interfaces Types
- Virtual routers and Layer 3 interfaces

Learning Objectives

After you complete this module, you should be able to:

- Describe firewall network segmentation components used to block threats
- Configure firewall security zones to implement network segmentation
- Configure tap interfaces to collect network traffic for later analysis
- Configure virtual wire interfaces to control network traffic traversing between two firewall interfaces
- Configure Layer 3 interfaces to control network traffic traversing Layer 3 networks
- Configure a virtual router to support Layer 3 interfaces





Security Zones overview

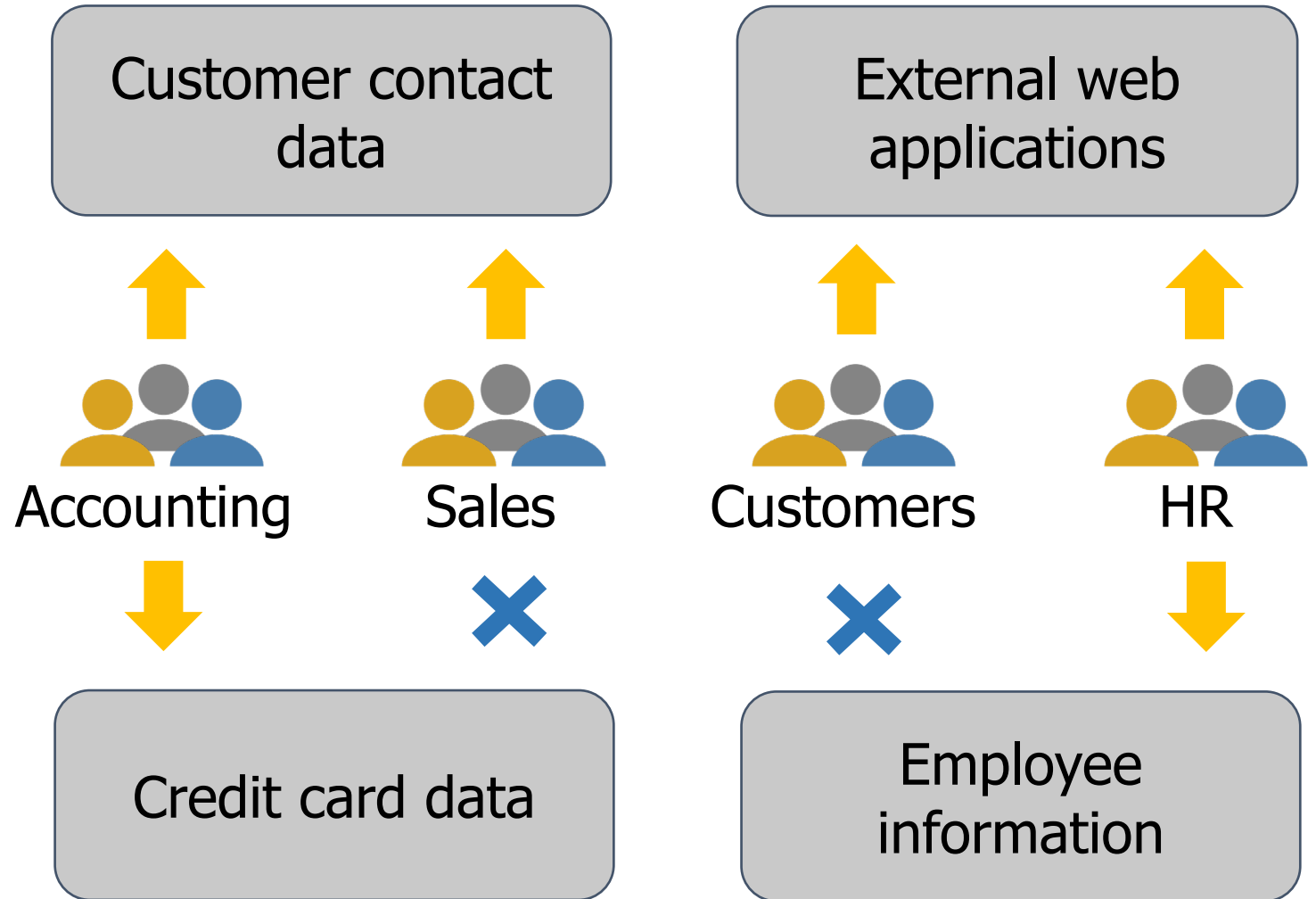
Network interfaces and security zones

Interfaces Types

Virtual routers and Layer 3 interfaces

Network Segmentation

- Use network segmentation to secure access to data.
- Understand your business and organizational drivers:
 - Who must access what?
 - Use the principle of least privilege.
 - Consider any regulatory requirements.



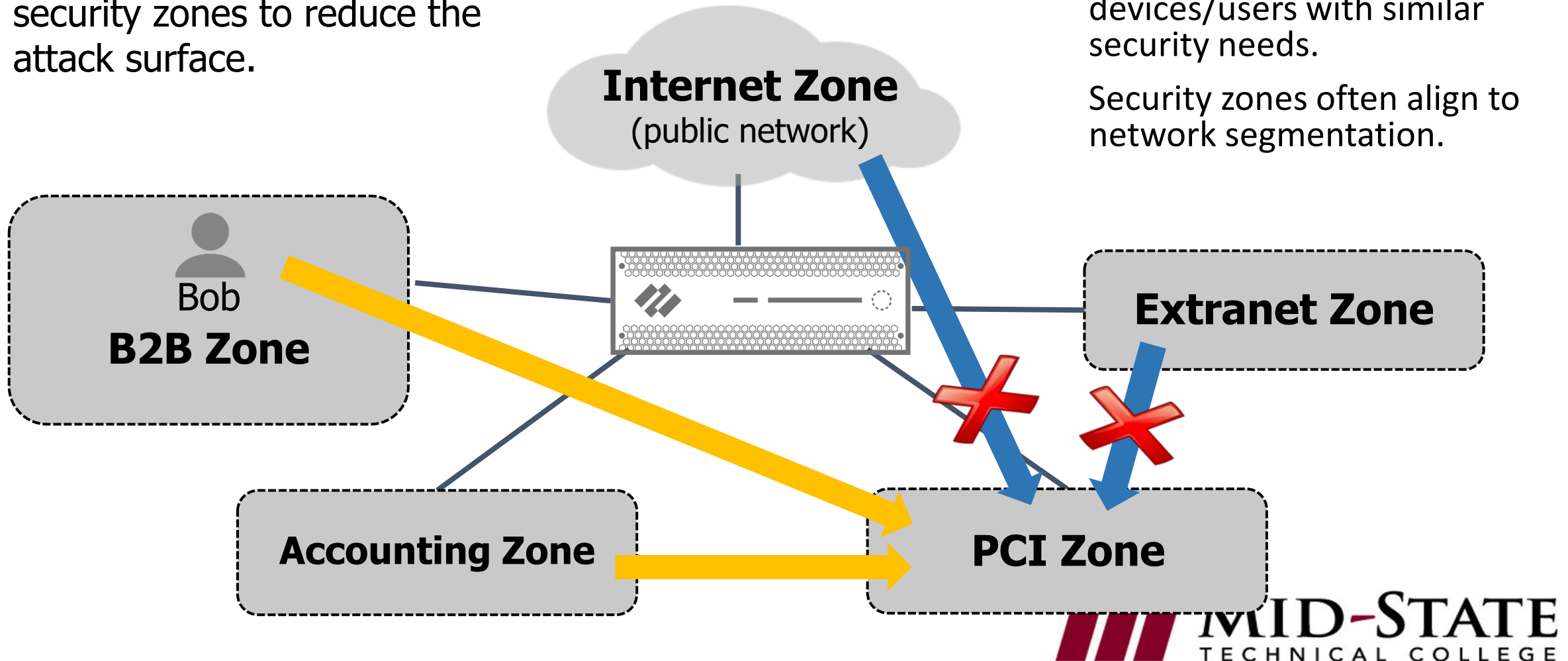
For a Tech Doc about this topic, log into Live and search for "Network Segmentation Using Zones."

Network Segmentation and Security Zones

Use network segmentation and security zones to reduce the attack surface.

Security zones group devices/users with similar security needs.

Security zones often align to network segmentation.



Configure Security Policy to Support Segmentation

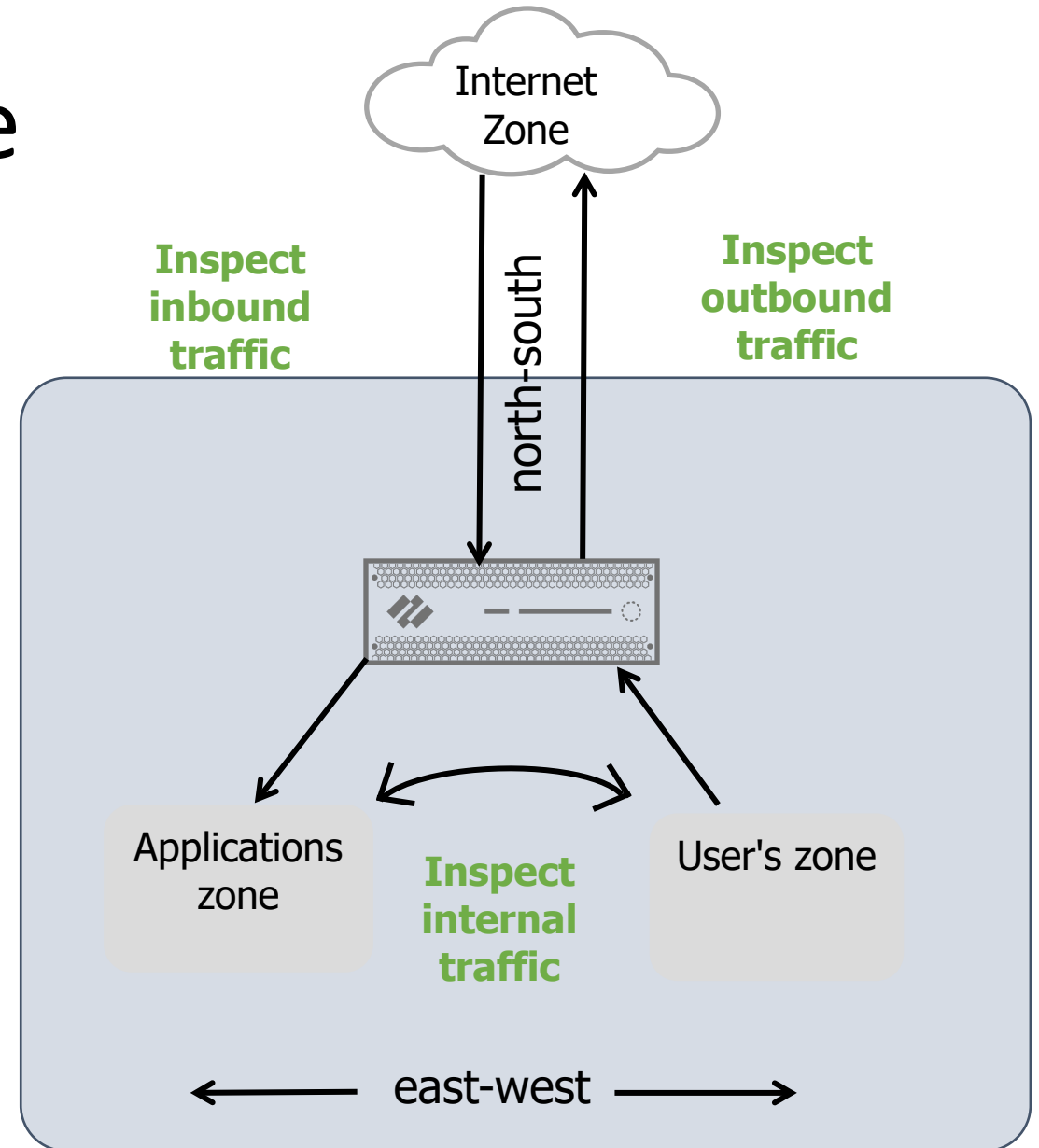
Policies > Security

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRE...	DEVICE			
1	B2B-PCI-Access	B2B	universal	B2B	any	Bob	any	PCI	any	any	mssql-db	application-default	Allow
2	Acct-PCI-Access	Accounting	universal	Accounting	any	Accounting_Grp	any	PCI	any	any	mssql-db	application-default	Allow
3	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	any	Allow
4	interzone-default	none	interzone	any	any	any	any	any	any	any	any	any	Deny

- Create a Security policy rule to allow required interzone traffic:
 - *Bob* in the *B2B* zone is allowed to access the *PCI* zone.
 - The *Accounting-Grp* in the *Accounting* zone is allowed to access the *PCI* zone.
- Any other interzone traffic is blocked, by default.

Zero Trust Architecture

- Never trust, always verify.
- Inspect perimeter traffic:
 - Inbound traffic
 - Outbound traffic
- Also inspect internal traffic.



For a Tech Doc about this topic, log into Live and search for "What is Zero Trust Architecture."

Security Zones overview



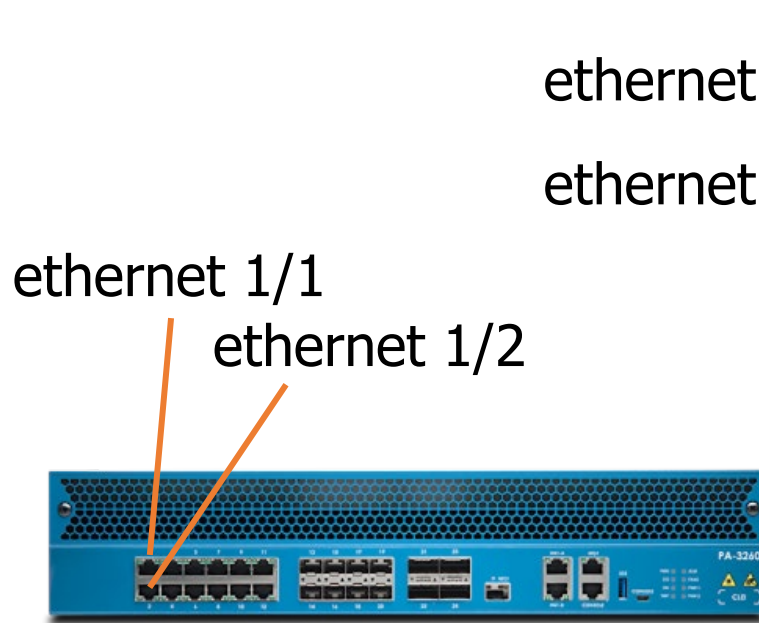
Network interfaces and security zones

Interfaces Types

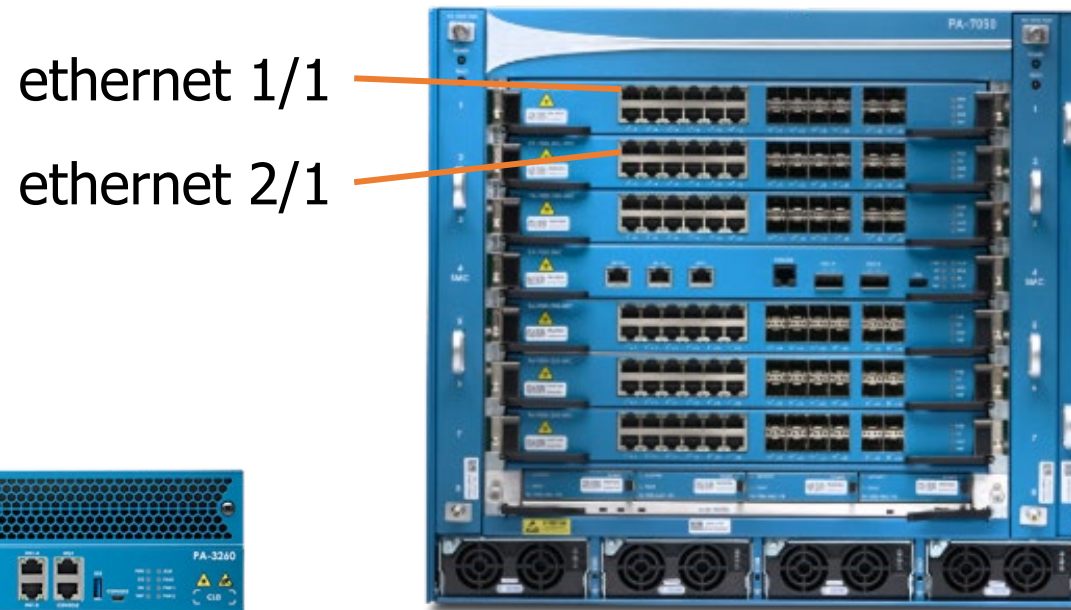
Virtual routers and Layer 3 interfaces

Network Interfaces

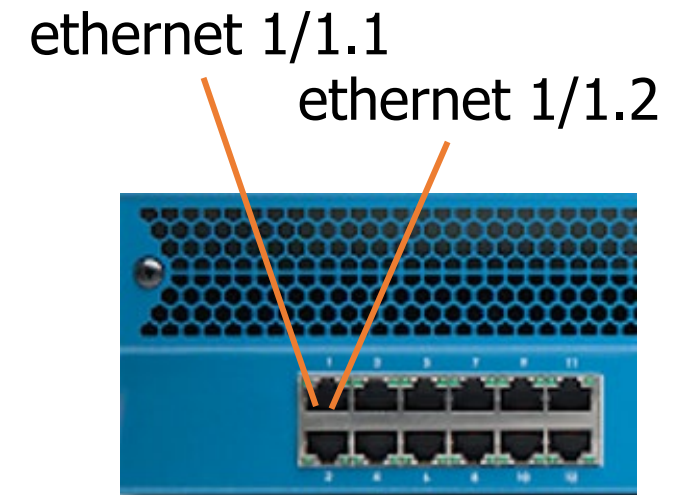
- The firewall data plane controls *in-band* network interfaces.
- Each interface is assigned to a single zone.
- A zone can include multiple physical or logical interfaces.



Single-Slot Firewall



Multi-Slot Firewall



Logical Interfaces

Interface Types and Zone Types

Different zone types support only specific interface types:

Tap Zone

Tap interfaces

Layer 2 Zone

Layer 2 interfaces

Layer 3 Zone

- Layer 3 interfaces
- VLAN interfaces
- Loopback interfaces
- Tunnel interfaces

Tunnel Zone

No interfaces assigned

Virtual Wire Zone

Virtual wire interfaces

MGT and HA interfaces are not assigned to a zone.

Create a Security Zone

Network > Zones > Add

- Specify zone **Name**.
- Specify zone **Type**.
- Assign **Interfaces**:
 - Must be appropriate type.
 - Unassigned interfaces do not process traffic.

Security Zones overview

Network interfaces and security zones

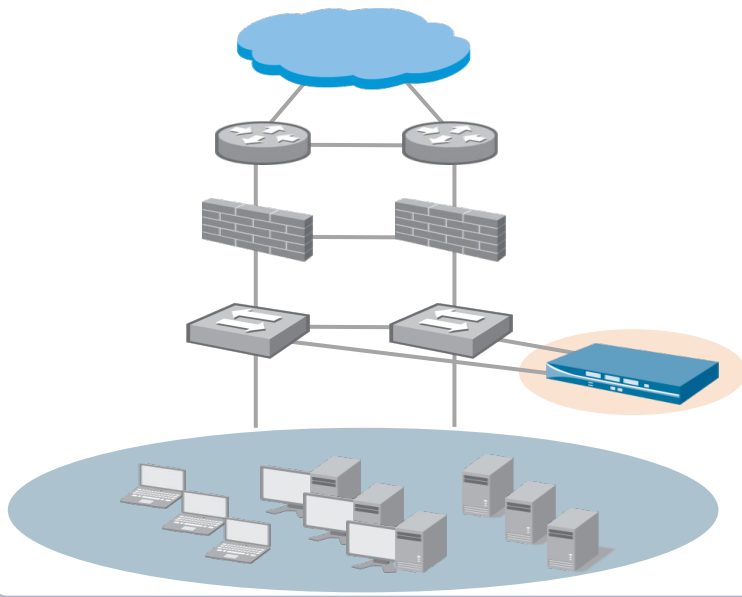
Interfaces Types



Virtual routers and Layer 3 interfaces

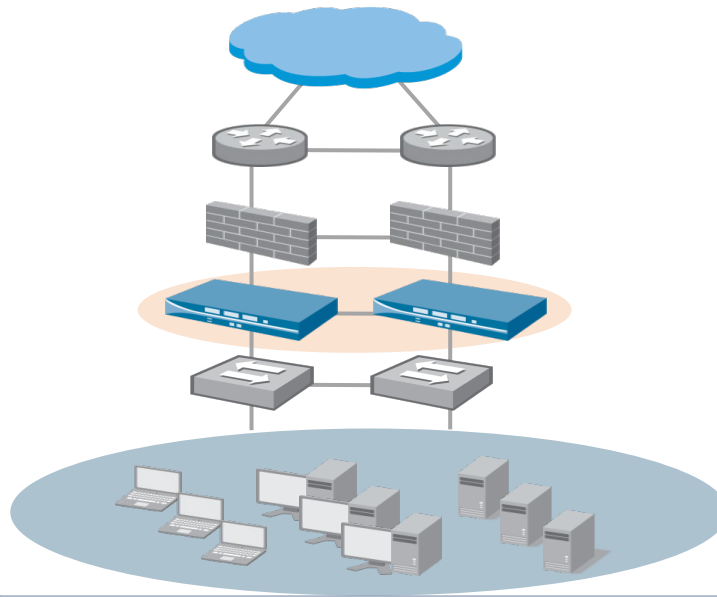
Flexible Deployment Options for Ethernet Interfaces

Tap



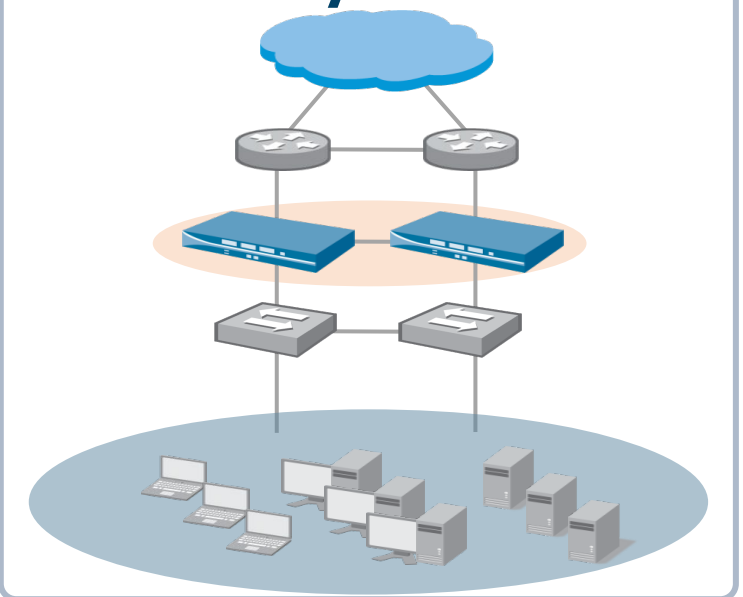
- Application, user, and content visibility without inline deployment
- Used for evaluation and audit of existing networks

Virtual Wire



- App-ID, Content-ID, User-ID, and SSL decryption
- Includes NAT capability

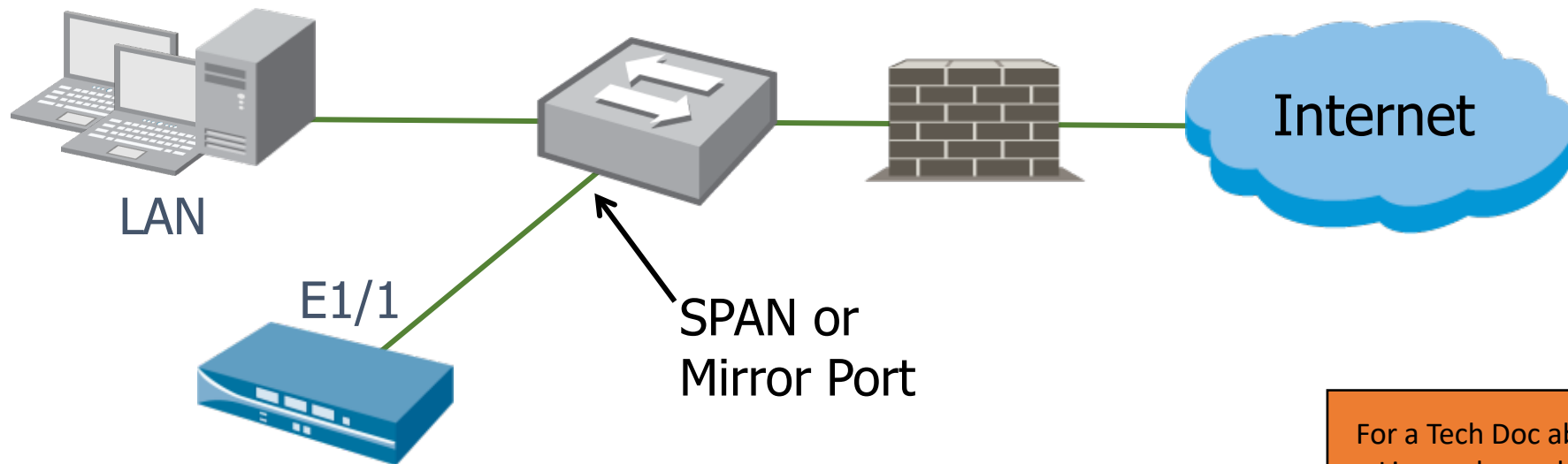
Layer 3



- All the virtual wire mode capabilities with the addition of Layer 3 services: virtual routers, VPN, and routing protocols

Tap Interfaces

- Enable passive monitoring of switch traffic from a SPAN or mirror port
- Cannot control traffic or perform traffic shaping
- Must be assigned to a tap zone
- Use Traffic log information to configure Security policy rules



For a Tech Doc about this topic, log into Live and search for “Tap Interfaces.”

Configure a Tap Interface

Network > Interfaces > Ethernet > <select_interface>

Ethernet | VLAN | Loopback | Tunnel | SD-WAN

Q

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE
ethernet1/1									
ethernet1/2									
ethernet1/3									
ethernet1/4									
ethernet1/5									
ethernet1/6									
ethernet1/7									
ethernet1/8									
ethernet1/9									

Ethernet Interface

Interface Name ethernet1/3

Comment Tap interface for monitoring traffic only

Interface Type Tap

Netflow Profile None

Config | Advanced

Assign Interface To

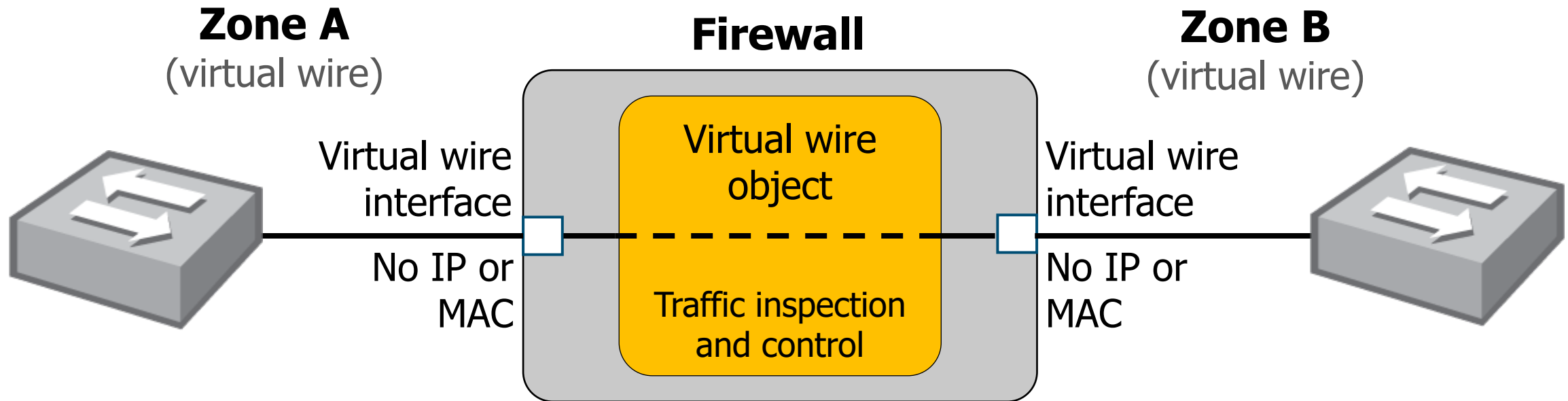
Security Zone Monitor_Only_Zone

Select **Tap** as the Interface Type.

Select a tap type **Security Zone**.

Virtual Wire Interfaces

- Bind two firewall interfaces together through a virtual wire object
- Typically used when no switching or routing is needed
- No configuration changes for adjacent network devices



Configure a Virtual Wire Object

- A virtual wire object connects to virtual wire interfaces.
- A virtual wire can accept traffic based on 802.1Q VLAN tags:
 - 0 = untagged traffic

Network > Virtual Wires > Add

The screenshot shows the 'Virtual Wire' configuration window. The title bar says 'Virtual Wire' with a help icon. The configuration fields are as follows:

Field	Value
Name	Vwire_Object
Interface1	ethernet1/4
Interface2	ethernet1/5
Tag Allowed	[0 - 4094]

Below the fields, there is a note: 'Enter either integers (e.g. 100) or ranges (e.g. 100-200) separated by commas. Integer values cannot exceed 4094.'

At the bottom, there are two checkboxes:

- ☐ Multicast Firewalling
- ☒ Link State Pass Through

Two callout boxes provide additional context:

- A callout box points to the 'Tag Allowed' field with the text: 'Forward only multicast-traffic matched to a Security policy rule (optional).'
- A callout box points to the 'Link State Pass Through' checkbox with the text: 'Link state is forwarded.'

Configure a Virtual Wire Interface

Network > Interfaces > Ethernet > <select_interface>

Ethernet Interface ?

Interface Name

ethernet1/5

Comment

Vwire for the Danger Zone

Interface Type

Virtual Wire

Netflow Profile

None

Config

Advanced

Assign Interface To

Virtual Wire

Vwire_Object

Security Zone

Danger

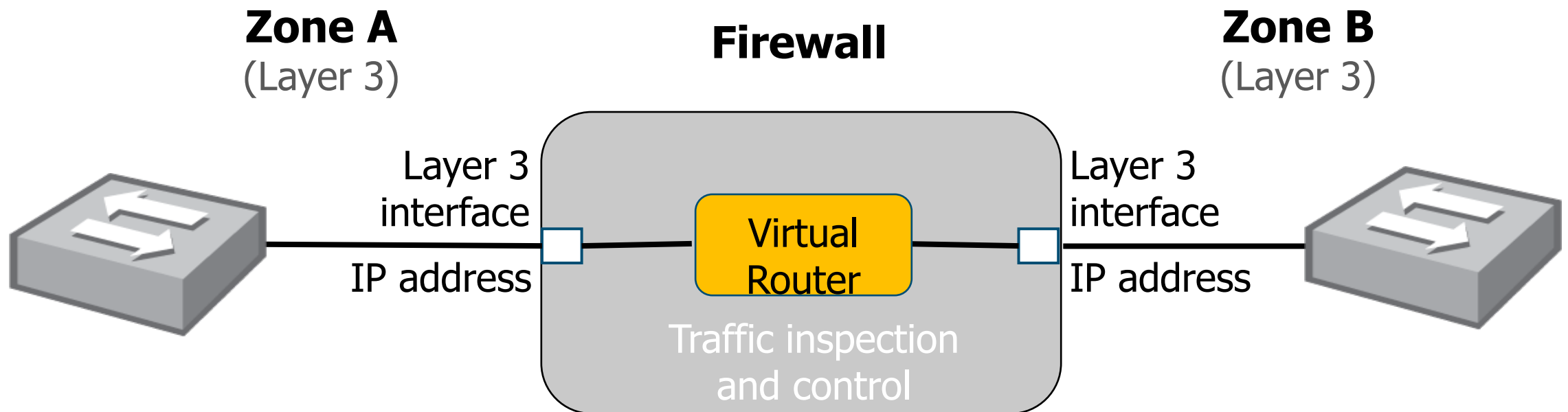
Select **Virtual Wire**.

Add virtual wire object now or later.

Select a virtual wire type **Security Zone**.

Layer 3 Interfaces

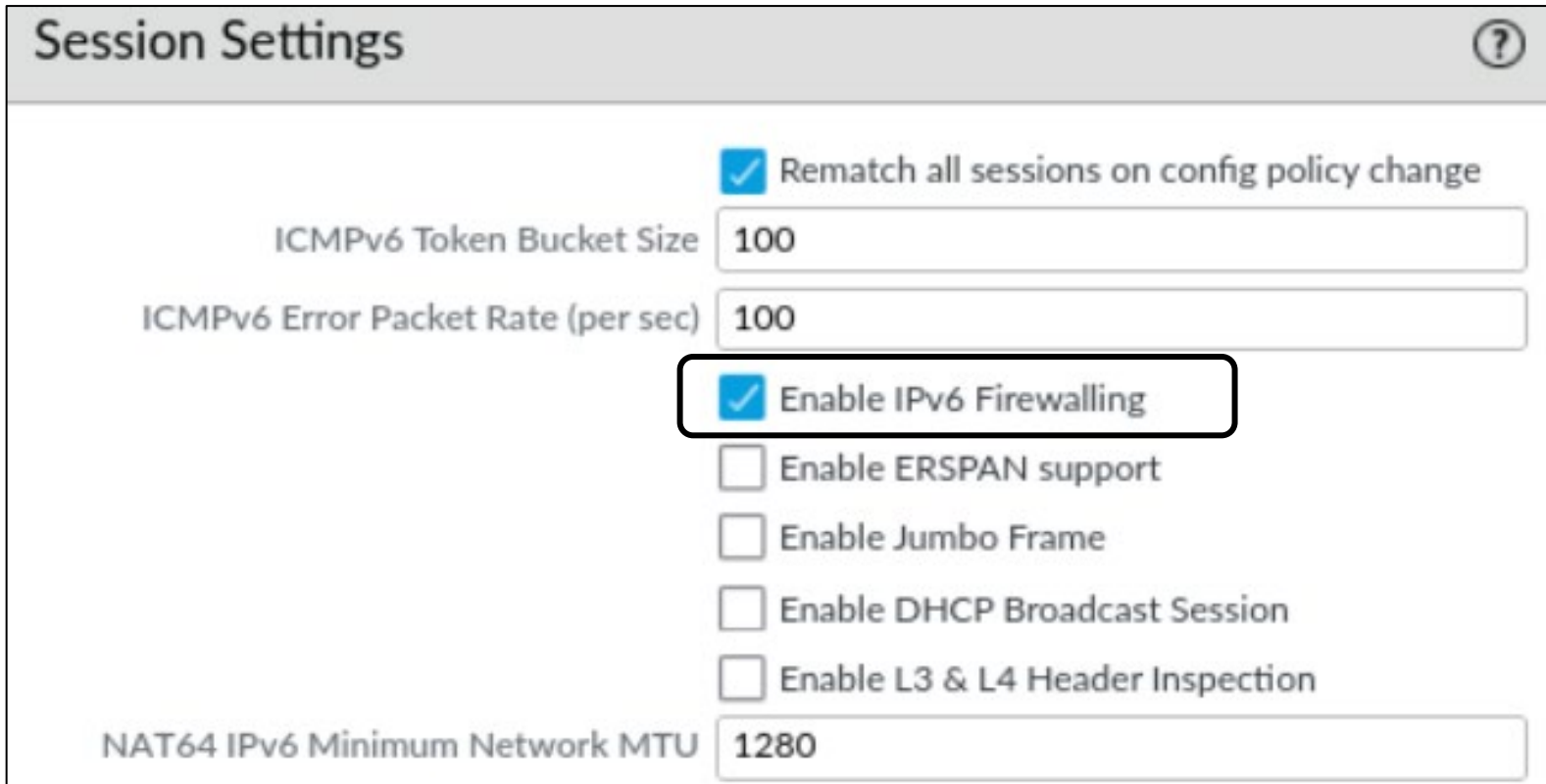
- Enables routing between multiple interfaces:
 - Requires a virtual router configuration
- Can require network configuration to accommodate new IP addresses



Enable IPv4 and IPv6 Support

- Layer 3 interfaces support IPv4 and IPv6.
- To support IPv6 addresses, you must enable IPv6 on the firewall.

Device > Setup > Session > Session Settings



Session Settings

☒ Rematch all sessions on config policy change

ICMPv6 Token Bucket Size

ICMPv6 Error Packet Rate (per sec)

☒ Enable IPv6 Firewalling

☐ Enable ERSPAN support

☐ Enable Jumbo Frame

☐ Enable DHCP Broadcast Session

☐ Enable L3 & L4 Header Inspection

NAT64 IPv6 Minimum Network MTU

Configure a Layer 3 Interface: Config

Network > Interfaces > Ethernet > <select_interface>

Ethernet Interface

Interface Name

ethernet1/1

Comment

Interface connected to the Internet

Interface Type

Layer3

Netflow Profile

None

Config

IPv4

IPv6

SD-WAN

Advanced

Assign Interface To

Virtual Router

VR-1

Security Zone

Internet

Select **Layer3**.

Select a **Virtual Router**.

Select a Layer 3 type **Security Zone**.

Configure a Layer 3 Interface: IPv4

Network > Interfaces > Ethernet > <select_interface>

Ethernet Interface

Config | **IPv4** | IPv6 | SD-WAN | Advanced

Interface Name: ethernet1/1
Comment: Interface connected to the
Interface Type: Layer3
Netflow Profile: None

Config | **IPv4** | IPv6 | SD-WAN

☐ Enable SD-WAN
Type: ☐ Static ☐ PPPoE ☒ DHCP Client
☒ Enable
☒ Automatically create default route pointing to default gateway provided by server
☐ Send Hostname: system-hostname
Default Route Metric: 10
[Show DHCP Client Runtime Info](#)

☐ IP
☐ 203.0.113.20/24

+ Add - Delete ↑ Move Up ↓ Move Down

IP address/netmask. Ex. 192.168.2.254/24

Select to specify a static or DHCP-assigned IP address.

Enter static IP addresses with CIDR notation.

STATE
L COLLEGE

Configure a Layer 3 Interface: Advanced

Network > Interfaces > Ethernet > <select_interface>

Ethernet Interface

Interface Name

Comment

Layer3

one

IPv6 | SD-WAN | **Advanced**

Link Speed Link Duplex Link State

Other Info | ARP Entries | ND Entries | NDP Proxy | LLDP | DDNS

Management Profile

MTU

☐ Adjust TCP MSS

IPv4 MSS Adjustment

IPv6 MSS Adjustment

☐ Untagged Subinterface

Specify firewall management services accessible on this interface.

(IPv4) Pre-load ARP cache entries.

(IPv6) Configure NDP proxy.

Enable and configure DDNS.

(IPv6) Pre-load ND cache entries.

Enable and configure LLDP.

Interface Management Profile

Network > Network Profiles > Interface Mgmt > Add

The screenshot shows the 'Interface Management Profile' configuration window. The title bar reads 'Interface Management Profile'. Below the title bar, the 'Name' field contains 'HTTPS_SSH_Ping_SNMP_RPs'. The configuration is divided into two main sections: 'Administrative Management Services' and 'Network Services'. In the 'Administrative Management Services' section, 'HTTP' is unchecked, 'HTTPS' is checked, 'Telnet' is unchecked, and 'SSH' is checked. A callout box labeled 'Web management interface access' points to the 'SSH' checkbox. In the 'Network Services' section, 'Ping' is checked, 'HTTP OCSP' is unchecked, 'SNMP' is checked, 'Response Pages' is checked, 'User-ID' is unchecked, 'User-ID Syslog Listener-SSL' is unchecked, and 'User-ID Syslog Listener-UDP' is unchecked. A callout box labeled 'Common interface services' points to the 'Response Pages' checkbox. To the right of these sections is a table titled 'PERMITTED IP ADDRESSES'. A callout box labeled 'Restrict access to only permitted addresses' points to this table. At the bottom of the window, there are '+ Add' and '- Delete' buttons, and a text box with examples: 'Ex. IPv4 192.168.1.1 or 192.168.1.0/24 or IPv6 2001:db8:123:1::1 or 2001:db8:123:1::/64'.

Interface Management Profile

Name: HTTPS_SSH_Ping_SNMP_RPs

Administrative Management Services

- ☐ HTTP
- ☒ HTTPS
- ☐ Telnet
- ☒ SSH

Network Services

- ☒ Ping
- ☐ HTTP OCSP
- ☒ SNMP
- ☒ Response Pages
- ☐ User-ID
- ☐ User-ID Syslog Listener-SSL
- ☐ User-ID Syslog Listener-UDP

PERMITTED IP ADDRESSES

+ Add - Delete

Ex. IPv4 192.168.1.1 or 192.168.1.0/24 or IPv6 2001:db8:123:1::1 or 2001:db8:123:1::/64

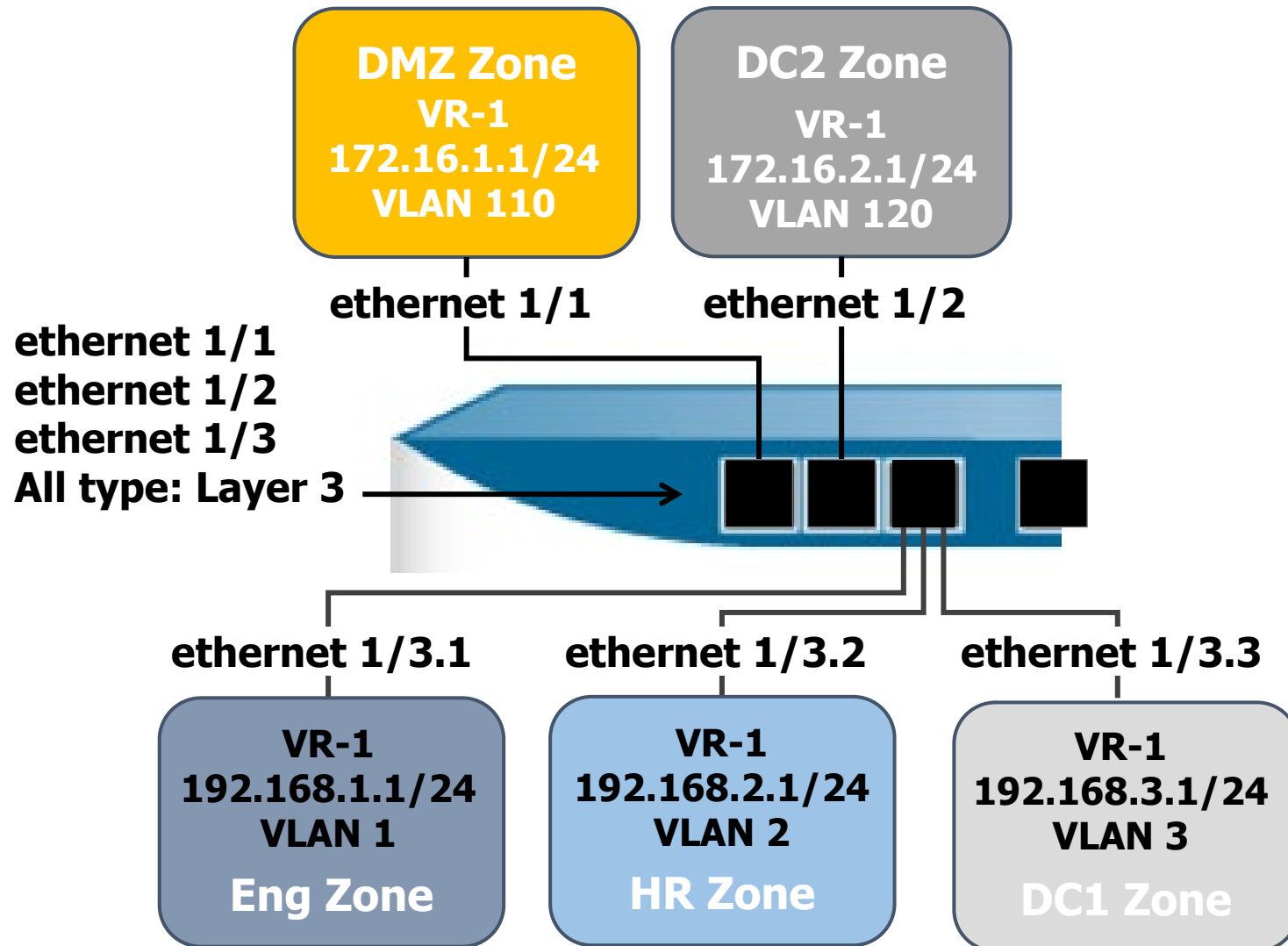
Web management interface access

Restrict access to only permitted addresses

Common interface services

- Defines which firewall management services are accessible from a traffic interface
- Can be applied to interfaces that support IP addresses:
 - Layer 3
 - Loopback
 - Tunnel

Layer 3 Subinterfaces



- Read and process traffic based on:
 - VLAN tags (1-4094)
 - VLAN tags and IP classifiers (source IP)
 - IP classifiers (untagged traffic, source IP)
- Common uses include:
 - More granular security rules
 - Logically splitting network traffic

For a Tech Doc about this topic, log into Live and search for "Layer 3 Subinterface."

Configure a Layer 3 Subinterface

Network > Interfaces > Ethernet

Ethernet VLAN Loopback Tunnel	
INTERFACE	INTERFACE TYPE
 ethernet1/1	Layer3
 ethernet1/2	Layer3
 ethernet1/3	Layer3
 ethernet1/4	Virtual Wire
 ethernet1/5	Virtual Wire
 ethernet1/6	
 ethernet1/7	
 ethernet1/8	
 ethernet1/9	
 Add Subinterface  Delete  PDF/CSV	

Layer3 Subinterface

Interface Name: ethernet1/2

Comment: Layer 3 Subinterface for

Tag: 1

Netflow Profile: None

Config | IPv4 | IPv6 | SD-WAN | Advanced

Assign Interface To

Virtual Router: VR

Security Zone: Users_Net

Annotations:

- 802.1Q VLAN tag (points to Tag: 1)
- Subinterface ID (points to the '2' in ethernet1/2)

Configure remaining options as normal Layer 3 interfaces.

Security Zones overview

Network interfaces and security zones

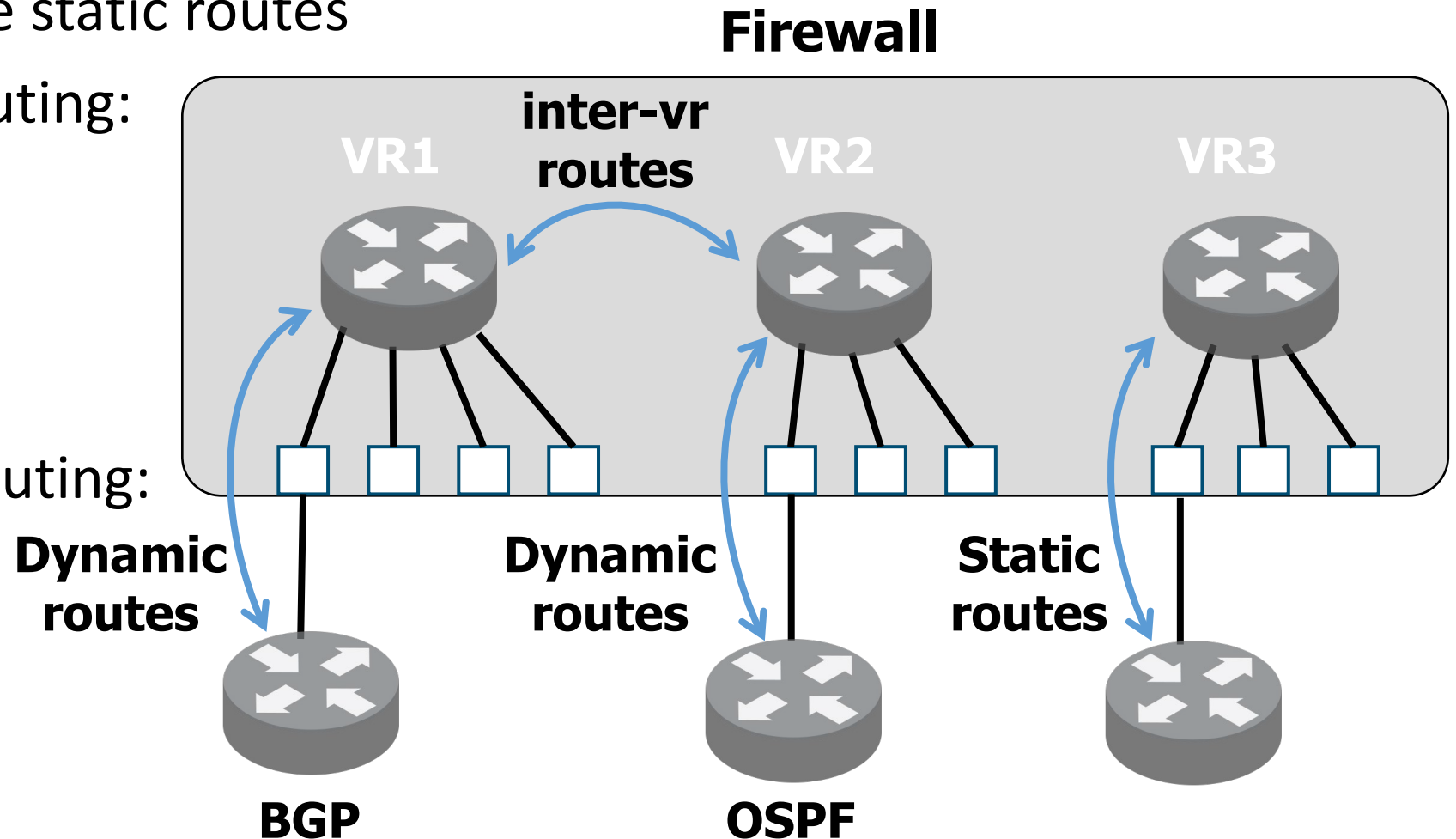
Interfaces Types

Virtual routers and Layer 3 interfaces



Virtual Routers

- Support one or more static routes
- Support dynamic routing:
 - BGPv4
 - OSPFv2
 - OSPFv3
 - RIPv2
- Support multicast routing:
 - PIM-SM
 - PIM-SSM



For a Tech Doc about this topic, log into Live and search for "Network > Virtual Routers."

Virtual Router General Settings

Network > Virtual Routers

Virtual Router - VR-1

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

BGP

Multicast

Name VR-1

General | ECMP

☐ INTERFACES ^

☐ ethernet1/1

☐ ethernet1/2

☐ ethernet1/3

☒ |

ethernet1/1

ethernet1/2

ethernet1/3

loopback

sdwan

tunnel

vlan

+ Add - Delete

Administrative Distances

Static 10

Static IPv6 10

OSPF Int 30

OSPF Ext 110

OSPFv3 Int 30

OSPFv3 Ext 110

IBGP 200

EBGP 20

RIP 120

Interfaces that the virtual router can use to forward traffic

Add a Static Default Route

Network > Virtual Routers > Static Routes > Add

Virtual Router - Static Route - IPv4

Name

Firewall-Default-Gateway

Destination

0.0.0.0/0

Interface

ethernet1/1

Next Hop

IP Address

203.0.113.1

Admin Distance

10 - 240

Metric

10

Route Table

Unicast

BFD Profile

Disable BFD

IP Address

Next VR

FQDN

Discard

None

Unicast

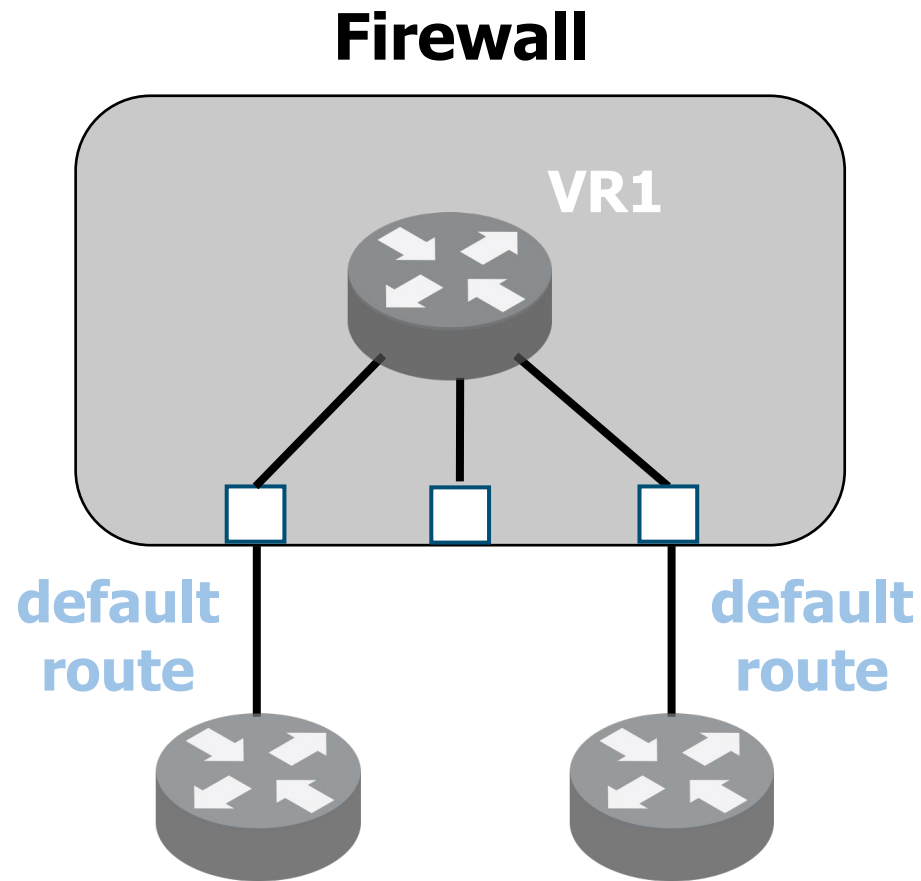
Multicast

Both

No Install

?

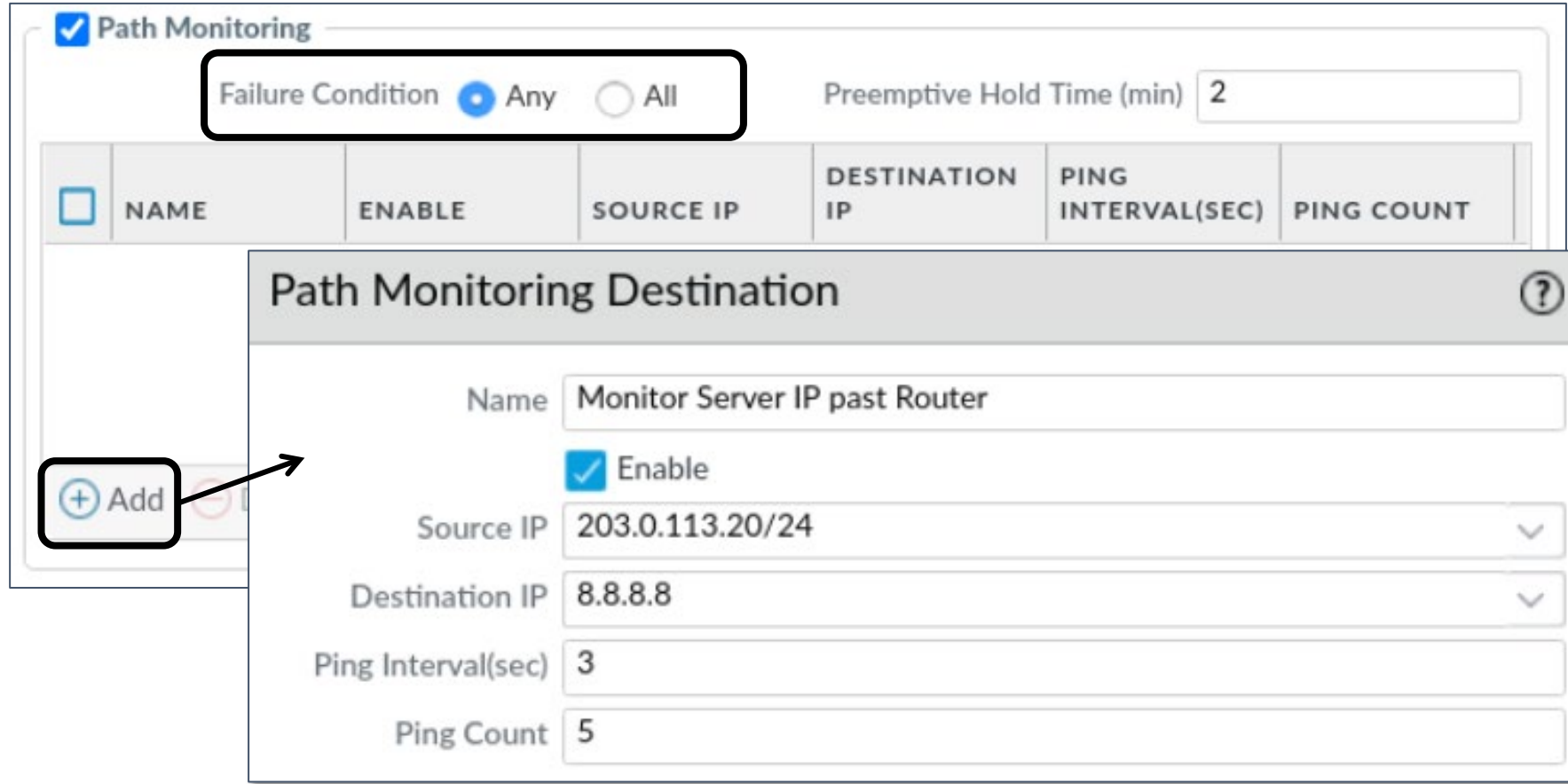
Multiple Static Default Routes



- Can configure multiple static default routes.
- Route with the lowest metric is used.
- Path monitoring determines if routes are usable.
- Firewall switches the default route during path failure.
- Supports failback.

Static Route Path Monitoring

Network > Virtual Routers > Static Routes > Add



☒ Path Monitoring

Failure Condition ☒ Any ☐ All

Preemptive Hold Time (min) 2

☐	NAME	ENABLE	SOURCE IP	DESTINATION IP	PING INTERVAL(SEC)	PING COUNT
--------------------------	------	--------	-----------	----------------	--------------------	------------

Path Monitoring Destination ⓘ

Name Monitor Server IP past Router

☒ Enable

Source IP 203.0.113.20/24

Destination IP 8.8.8.8

Ping Interval(sec) 3

Ping Count 5

- Uses ping to test reachability to stable upstream devices.
- Testing continues after failure.
- Will remove or re-add static routes.

Troubleshoot Routing

Network > Virtual Routers

☐	NAME	INTERFACES	CONFIGURATION	RIP	OSPF	OSPFV3	BGP	MULTICAST	RUNTIME STATS
☐	default		ECMP status: Disabled						More Runtime Stats
☐	VR-1	ethernet1/1	Static Routes: 1						More Runtime Stats

Virtual Router - VR-1

Routing

RIP

OSPF

OSPFv3

BGP

Multicast

BFD Summary Info

Route Table

Forwarding Table

Static Route Monitoring

Route Table

☒ Unicast

☐ Multicast

family

IPv4 and IPv6

7 items

→

×

	NEXT HOP	WEIGHT	FLAGS	AGE	INTERFACE
	203.0.113.0		A S		ethernet1/1
192.168.1.0/24	192.168.1.1		A C		ethernet1/2
192.168.1.1/32	0.0.0.0		A H		
192.168.50.0/24	192.168.50.1		A C		ethernet1/3
192.168.50.1/32	0.0.0.0	0	A H		
203.0.113.0/24	203.0.113.20	0	A C		ethernet1/1
203.0.113.20/32	0.0.0.0	0	A H		

Refresh

A:active, ?:loose, C:connect, H:host, S:static, ~:internal, R:rip, O:OSPF, B:bgp, Oi:OSPF intra-area, Oo:OSPF inter-area, O1:OSPF ext-ty

Status of monitored paths

All known routes (RIB)

Where traffic will be forwarded (FIB)

Module Summary

Now that you have completed this module, you should be able to:

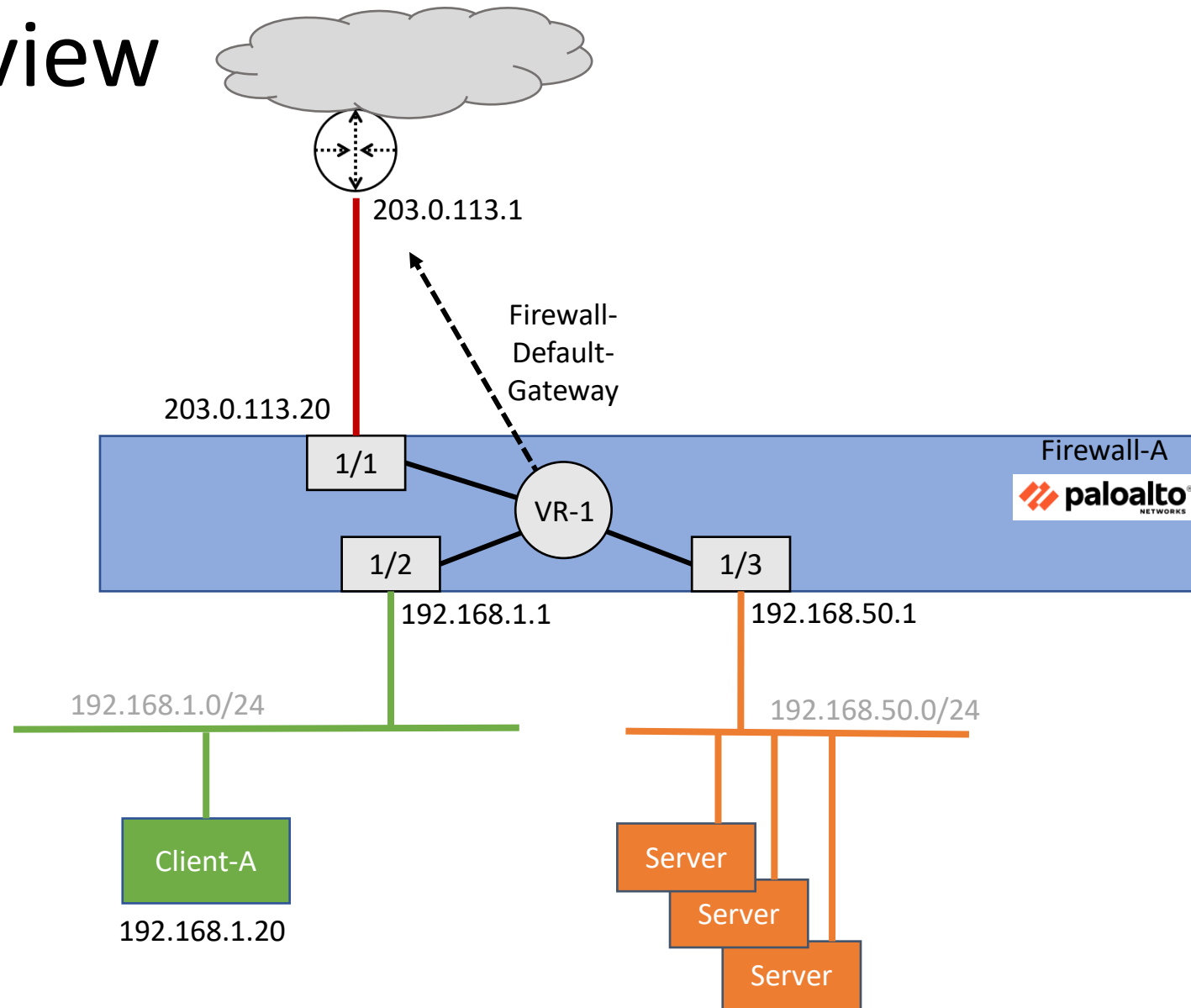
- Describe firewall network segmentation components used to block threats
- Configure firewall security zones to implement network segmentation
- Configure tap interfaces to collect network traffic for later analysis
- Configure virtual wire interfaces to control network traffic traversing between two firewall interfaces
- Configure Layer 3 interfaces to control network traffic traversing Layer 3 networks
- Configure a virtual router to support Layer 3 interfaces



Questions



Lab 5: Overview



Lab 5: Connecting the Firewall to Production Networks with Security Zones

- Load a baseline configuration
- Create Layer 3 interfaces
- Create a virtual router
- Segment your production network using security zones
- Test connectivity from firewall to hosts in each security zone
- Create Interface Management Profiles

End Content

Unit 5 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 5 Lab: Connecting the firewall to production networks with security zones
- Assignment 1: The State of Ransomware Memo
 - Choose 1 case study out of document (23pgs)